Retail Customer Shopping Behavior Analytics

Project Report

1. Project Title

Retail Customer Shopping Behavior Analytics

2. Introduction

Retail stores collect a lot of data every time a customer buys something. If this data is studied properly, a business can see who its customers are, what they buy, how they pay and how they react to offers such as discounts and subscriptions. This project analyzes a dataset of 3,900 retail customers to understand their shopping behavior and to find points that could help in business decisions.

3. Project Objectives

- Understand the purchasing patterns of customers.
- Analyze sales across the different product categories.
- Compare customer groups by gender and age group.
- Study discount and subscription behavior.
- Look at payment methods, shipping types and product ratings.
- Find useful business insights from the results.

4. Tools and Technologies Used

Python, Pandas, PostgreSQL, SQL, Matplotlib, Seaborn and Jupyter Notebook.

5. Methodology

1. Loaded the customer shopping dataset (3,900 rows and 18 columns) from a CSV file using Pandas and explored it.
2. Checked the data for missing values, duplicates and inconsistencies. Only the review rating column had missing values (37 rows), and there were no duplicate rows.
3. Cleaned the data: filled the 37 missing ratings with the median rating of the same product category, changed the column names to snake_case, created an age group column and a purchase frequency in days column (for example, Weekly = 7 days), and dropped the promo code column because it was identical to the discount column.
4. Connected Python to PostgreSQL using SQLAlchemy and saved the cleaned data in a table called customer.
5. Wrote SQL queries to answer 10 business questions: revenue by gender, spending levels, product categories, subscription, discounts, age groups, purchase frequency, ratings, shipping and payment, and repeat customers.
6. Used Pandas for further analysis, such as value counts, group totals and the correlation between rating and purchase amount.
7. Created charts with Matplotlib and Seaborn inside the Jupyter Notebook.
8. Studied the results and wrote down the main observations.

Note: the dataset has no date column, so trends over time could not be analyzed.

6. Key Observations and Findings

- Total revenue was $233,081 from 3,900 customers. The average purchase was $59.76 and the average rating was 3.75.
- Gender: male customers brought in $157,890 (67.7%) and female customers $75,191 (32.3%). The average purchase was almost the same ($59.54 for men and $60.25 for women), so the gap comes from the number of customers.
- Categories: Clothing had the highest revenue at $104,264 (44.7%), then Accessories (31.8%), Footwear (15.5%) and Outerwear (7.9%). Outerwear also had the lowest average purchase ($57.17).
- Spending levels: the 1,297 customers who spent $74 or more produced 48.5% of the revenue, while customers who spent less than $45 produced 17.0%.
- Subscription: 1,053 customers (27%) were subscribers and they produced 26.9% of the revenue. Their average purchase ($59.49) was close to that of non-subscribers ($59.87).
- Discounts: 43% of customers (1,677) had a discount. Their average purchase was $59.28, compared with $60.13 without a discount. The share of discounted purchases was similar in all categories (42.1% to 44.4%).
- Age groups: young adults (18 to 31) had the highest revenue ($62,143, 26.7%) and seniors (58 to 70) the lowest ($55,763, 23.9%). Middle-aged customers and young adults buy slightly more often (about every 84 days on average) than seniors (about every 97 days).
- Payment and shipping: PayPal (677 customers), Credit Card (671) and Cash (670) were used the most. Credit Card had the highest revenue ($40,310). Free Shipping had the highest revenue ($40,777) and Standard shipping the lowest average purchase ($58.46).
- Ratings: average ratings by category were between 3.72 and 3.79. Gloves had the best rating (3.86). The correlation between rating and purchase amount was 0.03, so there is no clear relation.
- Repeat customers: 97.9% of customers had more than one previous purchase, so this did not separate customers into useful groups.

Limitations: all female customers have no subscription and no discount, and all subscribers are male with a discount, so the effect of gender, subscription and discount cannot be separated. Most differences between groups are small.

7. Business Insights

The points below are possible recommendations based on the findings above, not proven results.

- Customer preferences: Clothing and Accessories bring most of the revenue, so these categories deserve the most attention in stocking and promotion.
- Category strategy: Outerwear has the smallest share and the lowest average purchase. The range and prices could be reviewed.
- Discount campaigns: discounted purchases were not larger than other purchases. Discounts could be tested on a small group before spending more on them.
- Retention: a third of customers produce nearly half of the revenue, so offers to keep these customers could be useful.
- Marketing: women spend about the same per purchase as men but are only a third of the customers, so campaigns aimed at women could be tried.

8. Conclusion

This project showed how a customer dataset can be turned into useful information. Python and Pandas were used to check and clean the data, and SQL on PostgreSQL made it easy to group the data and answer the business questions. Charts made in Matplotlib and Seaborn helped to see the differences between groups quickly. The main finding is that most of the revenue comes from male customers, from high-spending customers and from the Clothing and Accessories categories, while most other differences are small. More data, especially purchase dates, would allow deeper analysis in the future.